3.10 LORENTZ FORCE

When an electric charge q is kept at rest in a magnetic field, no force acts on it. At the same time, if the charge moves in the magnetic field, it experiences a force. This force is different from Coulomb force, studied in unit 1. This force is known as magnetic force. It is given by the equation

$$\vec{F} = q\left(\vec{v} \times \vec{B}\right) \qquad (3.54)$$

In general, if the charge is moving in both the electric and magnetic fields, the total force experienced by the charge is given by $\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$. It is known as Lorentz force.

3.10.1 Force on a moving charge in a magnetic field

When an electric charge q is moving with velocity $\vec{v}$ in the magnetic field $\vec{B}$, it experiences a force, called magnetic force $\vec{F}_m$. After careful experiments, Lorentz deduced the force experienced by a moving charge in the magnetic field $\vec{F}_m$

$$\vec{F}_m = q\left(\vec{v} \times \vec{B}\right) \qquad (3.55)$$

In magnitude, $F_m = qvB\sin\theta$ (3.56)

The equations (3.55) and (3.56) imply

1. $\vec{F}_m$ is directly proportional to the magnetic field $\vec{B}$
2. $\vec{F}_m$ is directly proportional to the velocity $\vec{v}$ of the moving charge
3. $\vec{F}_m$ is directly proportional to sine of the angle between the velocity and magnetic field
4. $\vec{F}_m$ is directly proportional to the magnitude of the charge q
5. The direction of $\vec{F}_m$ is always perpendicular to $\vec{v}$ and $\vec{B}$ as $\vec{F}_m$ is the cross product of $\vec{v}$ and $\vec{B}$

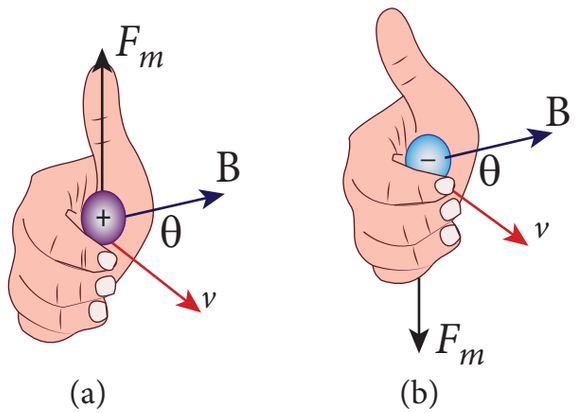


Figure 3.44 Direction of the Lorentz force on (a) positive charge (b) negative charge

6. The direction of $\vec{F}_m$ on negative charge is opposite to the direction of $\vec{F}_m$ on positive charge provided other factors are identical as shown Figure 3.44 (b)
7. If velocity $\vec{v}$ of the charge q is along magnetic field $\vec{B}$ then, $\vec{F}_m$ is zero

Definition of tesla

The strength of the magnetic field is one tesla if a unit charge moving in it with unit velocity experiences unit force.

$$1\ \mathrm{T} = \frac{1\,\mathrm{N\,s}}{\mathrm{C\,m}} = 1\frac{\mathrm{N}}{\mathrm{A\,m}} = 1\,\mathrm{N\,A^{-1}m^{-1}}$$

EXAMPLE 3.17

A particle of charge q moves with velocity $\vec{v}$ along positive y - direction in a magnetic field $\vec{B}$. Compute the Lorentz force experienced by the particle (a) when

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